Historical Oregon Floods

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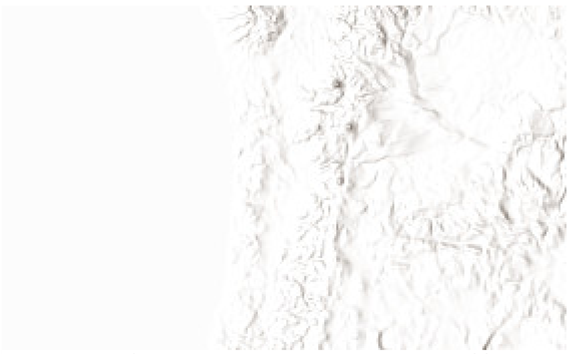
NWS Portland, OR

Draft

Introduction



Service Area

There is a long history of damaging floods in Northwest Oregon and Southwest Washington. This StoryMap addresses recent floods between 1964 and 2015 that caused widespread impacts to the region. Flooding in NWS Portland's area typically occurs between November and February. All of the floods documented here were caused by atmospheric rivers bringing heavy precipitation over a 1 to 3 day period. Some of these floods were prolonged by multiple periods of heavy rain, with flooding on some rivers occurring over multiple weeks.

February 1996

Summary:

This was the major flood event in recent history. Widespread flooding affected the entire Pacific Northwest and was especially notable in Western Oregon and Western Washington. Several tributaries in the Willamette Valley set new record stages. The flooding was the culmination of a series of unusual weather events. First, the fall and winter had above-normal precipitation, on the order of 125%, although snowpack was below normal. Second, in mid and late January, heavy snow fell in mid and high elevations of the Cascades and coastal mountains, and significant snow accumulated even at low elevations.

Third, the period of snow was followed by a deep freeze, with freezing rain and frozen ground. Fourth, the weather pattern changed dramatically in early February, with a strong atmospheric river bringing warm, moist air to the region, which resulted in heavy rain and rapid snowmelt. Some river basins had 4-day rainfall totals exceeding 20 inches combined with another 10+ inches of water equivalent in melted snow. Rivers rose very rapidly February 6th-9th, with smaller creeks and rivers cresting on the 7th and 8th and larger rivers cresting on the 9th and 10th. There were 8 fatalities in Northwest Oregon, along with 3 fatalities in Washington. Several of these were the result of people driving their cars into flooded areas and being swept away. In addition, a lady died when her house along the Sandy River was undermined and slid into the river, and a young girl in Scio was swept away when she went outside to get the newspaper. There was a major volunteer effort in downtown Portland to build a plywood and sandbag wall at the top of the seawall. President Bill Clinton visited Portland and other locations in the Pacific Northwest to learn more about the damage and flood fight effort. Total damages across the Pacific Northwest exceeded 1 billion dollars.

Digital Photos: City of Portland Archives

Hourly precipitation vs Stream Flow over time

[insert scale photo]

Methodology:

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Interesting Notes

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Severity map based on max crest height

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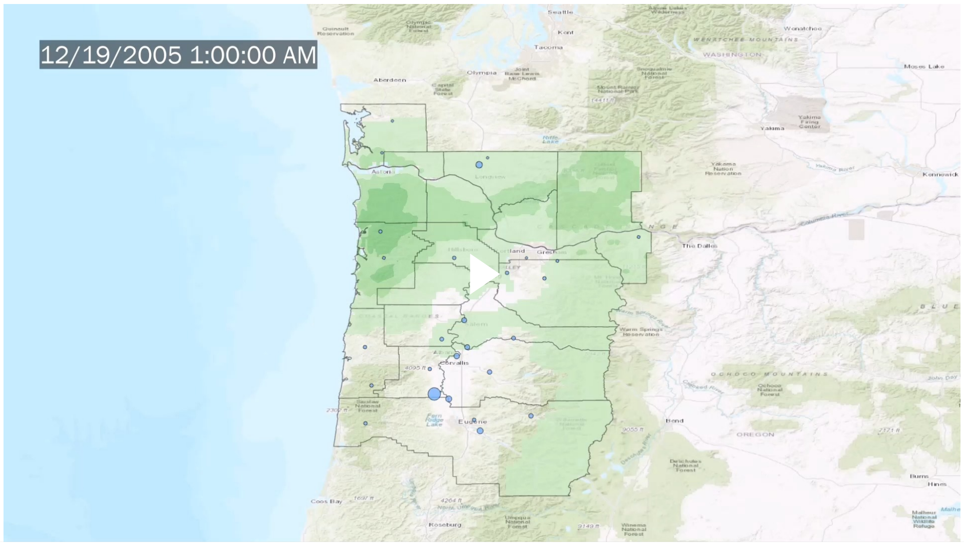
December 2005

Summary:

2 Primary Stages

Edit into 2 parts



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January 2006

Summary:

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November 2006

Summary:

After a dry summer and early fall, the wet season began abruptly on November 2nd, with short waves bringing periods of heavy rain through the 5th and quickly saturating the soil. Then, a very strong atmospheric river with deep tropical moisture brought record-setting heavy rain on the 6th and 7th, with many locations setting new 24-hr records, including a new Oregon record for 24 hour precipitation of 14.3” at Lees Camp and a new Washington record of 15.5” at June Lake SNOTEL. Record flooding occurred on the Wilson River in far Northwest Oregon on November 6th and 7th, causing nearly $15 million in damage in and around Tillamook. Near-major flooding on the lower Nehalem inundated parts of downtown Nehalem, causing about $1 million in damage. Other coastal rivers with minor flooding included the Siletz and Siuslaw. Inland, the heaviest rain occurred on the 7th. Significant damage was caused by flooding on the Sandy River and Hood River, which both drain from the slopes of Mt. Hood in the North Oregon Cascades. Flooding along the Sandy between Zigzag and Sandy caused major bank erosion and washed out parts of several houses and structures. In the Hood River basin, a debris flow on the White River inundated the Hwy 35 bridge with boulders and mud, and debris flows on the northeast side of Mt. Hood inundated other sections of Hwy 35 south of Parkdale. Flooding along Hood River also destroyed or damaged irrigation works. Dollar damage for the flooding on rivers draining Mt. Hood was near $40 million. Flooding also occurred on the Clackamas River east of Portland and on the Cowlitz and Lewis Rivers in southwest Washington. With all this rain in early November, numerous locations set new monthly precipitation records, including nearly 50" of rain at Laurel Mountain. Fortunately, no flood-related deaths or injuries occurred in Portland's HSA.

A home along the Sandy River is destroyed by the floodwaters

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December 2007

Summary:

A pair of strong storms in very rapid succession resulted in heavy rain, damaging wind, and major flooding for portions of NW Oregon and SW Washington. The first round of precipitation Dec 1-2 brought snow to higher elevations of the Coast Range, Willapa Hills, and Cascades, with heavy rain at lower elevations. As the center of low pressure shifted northward, an atmospheric river brought sub-tropical moisture and additional heavy rain Dec 2-4, with the axis of heaviest rainfall stretching southwest to northeast from Tillamook to Mt. St. Helens. Watersheds in the Willapa Hills and NW Oregon Coast Range had major, damaging flooding. Vernonia in Columbia County experienced damaging flooding for the first time since February 1996. Fishhawk Lake, a privately owned reservoir in western Columbia County.


Rockaway Beach, OR, 07 Dec 2007


Wheeler, OR, 07 Dec 2007


Centralia 4
Seattle 86

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January 2009

Summary:

Two distinct floods occurred, one on the 1st and 2nd and one on the 7th – 9th. For the event on the 1st/2nd, major flooding was driven by low-elevation snowmelt and heavy rain and occurred in the Portland area on Johnson Creek and the Clackamas River, with minor flooding on numerous coastal rivers and Willamette Valley tributaries. For the event on the 7th – 9th, the event was driven mainly by very heavy rain. Record flooding occurred on the Naselle River in Pacific County, and major flooding occurred on the Grays in Wahkiakum Co., the Willapa in Pacific Co., and the Nehalem in Tillamook Co. Significant flooding also occurred on the lower Cowlitz River, Lewis River, and other southwest Washington tributaries.


TILLAMOOK RV
REPAIR SALES
SPEED
25

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January 2012

Summary:

A complex storm system impacted Oregon January 17-21, 2012 and produced areas of heavy snow, damaging winds, heavy rainfall, significant flooding, landslides and freezing rain, resulting in a variety of weather-related impacts and damage in several Oregon counties. This weather system tapped into “atmospheric river” moisture over the Pacific and interacted with much colder air near the surface over portions of northern Oregon. The event was preceded by cold and relatively dry conditions the weekend of January 14th and 15th, with some rain and snow showers in the region. With cold air in place and a supply of warm, moist air transported to the region by a strong jet stream, the storm system produced 2 to 13 inches of wet snow at low elevations from late Tuesday evening (January 17), into Wednesday morning (January 18). The biggest snowfall accumulations at low elevations were in Columbia and Hood River Counties. The storm system also brought heavy rainfall to much of Oregon. Precipitation turned from snow to rain in the northern Willamette Valley and far-northwest Oregon during the morning of January 18th, with several hours of heavy rain following. Precipitation for valley locations south of the Portland metro area was in the form of rain throughout the event. A nearly stationary front slowly moved southward from northwest Oregon on the 18th, focusing heavy rain over much of west-central Oregon, where 4 to 15 inches fell. The snow level rose dramatically throughout the day on the 18th, resulting in heavy rain in the Coast Range and Cascade Foothills, while snow continued in the Cascades above 4000 feet. Precipitation rates diminished the evening of the 18th, but additional heavy rain occurred on the 19th.

The combination of melting snow and heavy rain resulted in sharp rises and significant flooding on numerous creeks and rivers. This storm system was noteworthy because of how quickly both the small streams and larger rivers rose. The biggest impacts from flooding occurred in west-central Oregon, where major flooding occurred along the Marys, Siuslaw, Luckiamute, and Alsea Rivers, along with Thomas Creek and Mill Creek near Salem. For some areas, it was the worst flooding since 1996. The Marys River in Benton County had a record flood, and the Luckiamute River in Benton and Polk Counties crested higher than in the February 1996 flood. Flooding along some slower-responding rivers, including the mainstem Willamette, continued through the 21st.

The heavy rainfall also produced several landslides that caused extensive road closures throughout the region. The counties impacted by landslides can provide specific information on impacts to roads and infrastructure. For example, it was reported that some homes in Benton County were destroyed by landslides.

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December 2015

Summary:

Multiple periods of heavy rainfall occurred over NWS Portland's area between December 6 and 23, with rainfall amounts of 13 to 29 inches in the Coast Range and Cascades and 6 to 12 inches in the lowlands. Eight locations in western Oregon had record daily rainfall totals on December 17. Portland International Airport had 25 consecutive days of rain during the month of December and had the second wettest day on record on December 7 when 2.67 inches fell during a 24 hour period. The cumulative effects of the heavy rainfall saturated the soils which led to landslides and flooding of creeks and rivers. One flood related fatality occurred on December 9 in Columbia County when an elderly couple drove around a barricade and into a flooded area. The Marys River, Wilson River and Johnson Creek all exceeded major flood level, which required evacuations. Preliminary data indicates Johnson Creek in Multnomah County recorded a flood of record on December 7, eclipsing the November 1996 mark by a few hundredths of an inch.

The several days of heavy rain saturated the soils and consequently resulted in dozens of landslides across western Oregon. The landslides resulted in several road closures, the collapse or buckling of road surfaces in many locations and erosion of critical infrastructure. There was also one fatality in western Lane County from a fast moving landslide that plowed into a home in the early morning hours of December 18. A slow moving landslide in Clackamas County prompted evacuation of a 70 unit apartment complex in the days before Christmas.





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December 1964

Summary:

Successive heavy rainfall events of unprecedented intensity mixed with frozen topsoil from cold temperatures earlier that December led to rapid run-off. Intense warm rains from 21-23 destroyed the entire Pacific Northwest—damages set records unheard of since the 19th century. Confluent flow from cold arctic air and warm tropical air brought the first storms on the 18th, which began as snow. On average, snow levels hit 1 ft. in the coastal range, and in the Cascades, they were measured to be from 3-4 ft. On the evening of the 19th, snow switched to rain. Six days of consecutive storms swept most of the snowpack down valleys and into population areas. This led to record-setting levels of run-off in coastal streams and cascade basins. Later that January, a second period of high flows was concentrated in coastal streams. Most peaked on the 19th, 20th, and 25th.
This "100-year" flood killed 20 people throughout the pacific northwest while also destroying entire ports, hundreds of miles of roads, bridges, and homes, leading to thousands of Oregonians becoming homeless. The storm hit Oregon harder than any other state and accrued approximately 158 million dollars in damages. Accounting for inflation, this would be 1.5 billion dollars in today's money.




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Damages

Combined Crest Height

Dec 1964

Feb 1996

Dec 2005

Jan 2006

Nov 2006

Dec 2007

2009

2012

2015